

Artificial Intelligence

Multi-Robot Path Planning on a Grid

Assignment-1

Submitted by

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1. State Representation

To accurately track each robot's progress through a dynamic environment, the search algorithm utilizes a custom State class. A simple (x, y) coordinate is insufficient because the validity of a cell changes over time due to other robots' reservations, and the robot must visit specific checkpoints in order.

Therefore, the state is represented by a tuple of four variables: (x, y, cp_ind, t):

- **x and y:** The current grid coordinates of the robot.
- **cp_ind (Checkpoint Index):** An integer tracking which checkpoint the robot is currently navigating towards. It starts at 0 and increments as each checkpoint is reached.
- **t (Time/Energy):** The current time step. This is crucial for checking against the space-time reservations set to avoid collisions with higher-priority robots.

The state also stores a parent pointer to reconstruct the final path once the goal is reached.

2. Search Algorithm Used

The algorithm implemented for this assignment is **Breadth-First Search (BFS)**.

Because every allowed action (moving to an adjacent cell or waiting in place) advances time by exactly 1 step and costs exactly 1 unit of energy, the pathfinding problem is effectively an unweighted graph search. BFS explores the state space level by level (which translates directly to time step by time step). This naturally guarantees that the first time the algorithm generates a state matching the goal coordinates and the final checkpoint index, it is the optimal path requiring the minimum time and energy.

Multi-robot coordination is handled sequentially via Priority-Based Planning . The highest-priority robot runs BFS first on an empty reservation set. Subsequent robots run BFS while treating the paths of previously planned robots as temporary obstacles in space-time (x, y, t).

3. Heuristic Function

No heuristic function was used for this implementation. Because BFS is an uninformed search algorithm, it blindly explores all valid neighbors level by level. While an informed search algorithm like A* could have been implemented using a Manhattan Distance heuristic (factoring in the sequence of remaining checkpoints), BFS is mathematically sufficient and fully justifiable for finding the optimal unweighted path as required by the assignment guidelines.